AGENTIC FRACTURE DETECTION: A MULTI-AGENT FUSIONOF FASTER R-CNN AND VISION-LANGUAGE MODELS

A PROJECT REPORT

Submitted by

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in partial fulfillment for the award of the degree of

BACHELOR OF ENGINEERING

in

COMPUTER SCIENCE AND ENGINEERING

RATHINAM TECHNICAL CAMPUS, COIMBATORE (AUTONOMOUS)

ANNA UNIVERSITY: CHENNAI-600 025

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BONAFIDE CERTIFICATE

Certified that this project report "**AGENTIC FRACTURE DETECTION: A MULTI-AGENT FUSION OF FASTER R-CNN AND VISION-LANGUAGE MODELS**" is the bonafide work of " **CHANDRU R S (721821104012), DHINESH KUMAR S (721821104017), GURUSANJAY KUBER M (721821104020), HARI BALAJI J (721821104021)**" who carried out the project work under my supervision during the **Academic year 2024-2025**.

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DECLARATION

We, Chandru R S, Dhinesh Kumar S, Gurusanjay Kuber M, Hari Balaji J hereby declare that the project report titled "**AGENTIC FRACTURE DETECTION: A MULTI-AGENT FUSION OF FASTER R-CNN AND VISION-LANGUAGE MODELS**" done by me under the guidance of **Mr. D. Shankar** at **Rathinam Technical Campus** is submitted in partial fulfilment of the requirements for the award of Bachelor of Engineering degree in Computer Science and Engineering. Certified further that, to the best of my knowledge, the work reported herein does not form part of any other project report or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

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ABSTRACT

The Bone fracture detection from X-ray images is an essential task in medical diagnosis, often requiring expert analysis. This paper introduces Agentic Fracture Detection (AFD), a novel approach that combines Faster R-CNN for fracture segmentation with vision-language models (VLMs) for explanation and decision support. Our system consists of four AI agents: (1) Perception Agent, which detects fractures using Faster R-CNN; (2) Reasoning Agent, which assesses fracture severity; (3) Interpretation Agent, which generates human-readable reports using a vision-language model; and (4) Decision Agent, which suggests next steps for diagnosis and treatment. Experimental results show improved accuracy in fracture detection and provide interactive, AI-assisted reports for radiologists. This approach enhances medical imaging analysis, making fracture detection more accurate, explainable, and assistive for healthcare professionals.

Keywords: *Fracture Segmentation, Agentic AI, Multi agent, Faster R-CNN, Vision Language Models(VLMs).*

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LIST OF ABBREVIATIONS

AI	-	Artificial Intelligence
NLP	-	Natural Language Processing
ML	-	Machine Learning
AFD	-	Agentic Fracture Detection
API	-	Application Programming Interface
CT	-	Computed Tomography
VLM	-	Visual Language Models
CNN	-	Convolutional Neural Network

CHAPTER 1

INTRODUCTION

1.1 Introduction

The rapid advancements in artificial intelligence (AI) and deep learning have significantly impacted the field of medical imaging, particularly in the detection and diagnosis of fractures. Traditional methods of fracture detection rely on radiologists manually analyzing X-ray images, which can be time-consuming and subject to human error. With the increasing workload on healthcare professionals, there is a pressing need for automated, accurate, and interpretable AI-driven solutions. This report introduces the development of an AI-based multi-agent system, Agentic Fracture Detection (AFD), which integrates Faster R-CNN and Vision-Language Models (VLMs) to improve the accuracy and explainability of bone fracture detection.

The primary motivation behind this project is to bridge the gap between AI's capability to detect fractures and the need for human-readable, decision-supportive outputs for radiologists. While existing AI models such as convolutional neural networks (CNNs) and object detection frameworks like Faster R-CNN, Mask R-CNN, and YOLO have demonstrated high accuracy in fracture classification, they often lack explainability. Moreover, these models operate as single-agent systems, limiting their ability to specialize in different aspects of the diagnostic process. The proposed AFD system adopts a multi-agent approach, where different AI agents collaborate to detect, interpret, and provide actionable insights for fracture diagnosis and treatment recommendations.

The AFD system comprises four specialized agents: the Perception Agent, Reasoning Agent, Interpretation Agent, and Decision Agent. The Perception Agent employs Faster R-CNN for accurate fracture localization, while the Reasoning Agent assesses the severity of the detected fractures. The Interpretation Agent utilizes VLMs to generate human-readable reports, ensuring that AI-driven analyses are interpretable by medical professionals. Finally, the Decision Agent provides recommendations for further diagnosis and treatment, enhancing the overall decision-making process for radiologists. This structured approach allows for improved specialization, higher diagnostic accuracy, and better integration with clinical workflows.

Another key innovation of the AFD system is its utilization of vision-language models, which enable the generation of detailed and interactive medical reports. Traditional AI-driven fracture detection systems provide only visual outputs, making it difficult for radiologists to interpret results. By incorporating VLMs, the AFD system enhances explainability, making it more aligned with the needs of medical professionals. Additionally, the system's ability to generate structured reports improves communication between AI systems and radiologists, fostering a more effective diagnostic process.

By implementing a multi-agent AI framework, the AFD system enhances the efficiency and accuracy of fracture detection while addressing key limitations found in existing methodologies. This collaborative approach ensures that each AI agent is specialized for a specific function, reducing computational complexity and improving diagnostic reliability. The integration of VLMs also facilitates a more user-friendly experience for medical practitioners, making AI-driven fracture detection more accessible and practical in real-world applications.

Moreover, the adoption of an agentic AI system in medical imaging highlights the broader potential of multi-agent architectures in healthcare. Beyond fracture detection, similar frameworks can be applied to other medical domains such as tumor detection, disease progression monitoring, and personalized treatment planning. This project serves as a pioneering step towards the future of AI-driven medical diagnostics, demonstrating the power of combining deep learning, vision-language integration, and structured AI collaboration for enhanced clinical decision-making.

The current state of AI in healthcare is rapidly evolving, with numerous research efforts focused on developing intelligent systems that can assist medical professionals. However, one of the major challenges remains the interpretability and trustworthiness of AI predictions. Many black-box AI models struggle to provide explanations that align with clinical reasoning, leading to skepticism among radiologists and healthcare providers. The AFD system addresses this concern by leveraging a structured, agent-based approach that enhances transparency, ensuring that AI-generated diagnoses and recommendations can be verified and trusted by experts.

The integration of AI with medical imaging also brings ethical and regulatory considerations. Deploying AI models in clinical settings requires adherence to strict

regulatory standards to ensure patient safety and data privacy. The AFD system incorporates explainability and fairness measures to mitigate biases and promote ethical AI usage. By ensuring that AI-driven decisions are interpretable and aligned with medical guidelines, the system aims to foster trust and facilitate seamless adoption in healthcare institutions.

From a technical standpoint, the implementation of a multi-agent framework in AI-driven diagnostics opens up new possibilities for enhanced performance and efficiency. Each agent in the AFD system is optimized for a specific task, reducing computational overhead and enabling real-time processing of medical images. This modular design allows for scalability, where additional agents can be introduced to handle more complex medical conditions beyond fractures, such as detecting soft tissue injuries or predicting patient recovery timelines.

The application of AI-driven fracture detection is particularly beneficial in resource-limited settings where access to expert radiologists is scarce. Automated systems like AFD can assist medical practitioners in rural or underprivileged areas by providing accurate and timely diagnoses, reducing the burden on healthcare facilities. Furthermore, integrating AI-powered solutions with telemedicine platforms can improve accessibility and enable remote consultations, ensuring that patients receive appropriate care regardless of geographical constraints.

1.2. Problem Statement

Medical imaging plays a crucial role in diagnosing bone fractures, which are among the most common injuries treated in healthcare facilities worldwide. Fractures can range from simple breaks that require minimal intervention to complex fractures that necessitate surgical procedures. Early and accurate diagnosis is critical in ensuring optimal treatment and preventing complications such as improper bone healing, chronic pain, and mobility issues. Traditionally, X-ray images are analyzed by radiologists to detect fractures, assess their severity, and recommend treatment. However, this manual approach is time-consuming, prone to human error, and highly dependent on the expertise of the radiologist. Misdiagnosis or delayed diagnosis can lead to improper treatment, prolonged recovery times, and severe complications.

The growing demand for automated fracture detection systems has led to advancements in artificial intelligence (AI)-based diagnostic tools. AI-powered

solutions have the potential to significantly improve accuracy and efficiency in medical imaging analysis. However, existing AI models still face challenges in accuracy, interpretability, and adaptability across different imaging datasets. The need for a robust, explainable, and scalable AI-based fracture detection system has become more pressing than ever, particularly in areas with limited access to trained medical professionals. AI-driven solutions must not only detect fractures but also provide insights into their severity and recommend treatment pathways, ensuring a comprehensive diagnostic approach.

Furthermore, as healthcare systems worldwide are overburdened with increasing patient loads, automating the diagnostic process for fractures can significantly reduce the workload on radiologists and healthcare providers. With the rise of deep learning and computer vision, leveraging sophisticated AI models that analyze X-ray images and provide reliable assessments can help optimize healthcare resources. However, achieving an optimal AI-driven solution requires addressing various technical and practical challenges, such as enhancing detection precision, improving model interpretability, and ensuring its seamless integration into clinical workflows. Additionally, AI-driven solutions must be designed to function effectively across diverse healthcare infrastructures, catering to hospitals with varying levels of technological adoption.

In many regions, there is a shortage of experienced radiologists, leading to increased diagnosis time and delayed treatment. This issue is especially prevalent in rural areas and developing countries, where access to specialized medical care is often limited. AI-based diagnostic tools could help bridge this gap by offering preliminary assessments that assist radiologists in prioritizing urgent cases. However, implementing AI models effectively requires extensive validation and compliance with regulatory standards to ensure patient safety and reliability. AI systems must be continuously refined to maintain diagnostic accuracy and reliability across different patient demographics and imaging conditions.

Differences in imaging techniques, equipment, and patient positioning can affect the clarity of X-ray images, making detection more difficult. Additionally, factors such as poor contrast, overlapping anatomical structures, and motion artifacts can further complicate the analysis, increasing the likelihood of diagnostic errors. A robust AI

model must be capable of handling such variability and extracting meaningful information from low-quality images. Advanced preprocessing techniques, such as noise reduction and image enhancement, may be necessary to improve AI model performance. Incorporating adaptive learning mechanisms can also enable AI models to improve their performance across different imaging conditions dynamically.

Subtle fractures, hairline fractures, and complex fractures can be difficult to detect, leading to false negatives and inappropriate treatments. Fractures in regions such as the wrist, spine, or hip may be particularly challenging to identify, even for trained radiologists. In some cases, fractures may be misclassified as normal bone variations or overlooked entirely, resulting in delayed treatment and worsening patient outcomes. AI-based fracture detection systems must demonstrate high sensitivity and specificity to minimize such diagnostic errors. Continuous model training using diverse datasets and real-world cases is crucial for enhancing the robustness of these systems. Additionally, incorporating multi-modal imaging data, such as CT scans and MRIs, can improve diagnostic precision by providing more comprehensive insights into bone structures and fracture patterns.

Many AI-based models provide predictions without clear reasoning, making it difficult for medical professionals to trust or validate the results. A black-box AI model that merely outputs a diagnosis without justification does not align with the needs of medical practitioners. Instead, AI models should offer explainable insights, highlighting the affected regions, quantifying fracture severity, and providing confidence scores for each prediction. This transparency is crucial for integrating AI into clinical workflows and ensuring that medical professionals can rely on its recommendations. The implementation of attention mechanisms, heatmaps, or decision trees within AI models can improve explainability and trustworthiness. Furthermore, AI-generated diagnostic reports should be aligned with medical terminology to facilitate seamless communication between AI systems and healthcare professionals.

Traditional AI models often require high computational power and may not be suitable for real-time diagnosis in clinical settings. Hospitals and diagnostic centers with limited computational infrastructure may struggle to implement sophisticated deep-learning models. Developing an optimized AI model with lightweight architectures

and efficient inference mechanisms is essential for widespread adoption, particularly in resource-constrained settings. The integration of edge computing and cloud-based AI solutions can help balance computational efficiency and accessibility, allowing remote diagnostic centers to leverage AI-driven fracture detection. Efficient AI architectures such as MobileNet and model quantization techniques can also help in reducing computational overhead while maintaining accuracy.

AI models trained on limited datasets may struggle to generalize across diverse patient populations. Factors such as age, bone density, ethnicity, and underlying medical conditions can impact fracture presentation in X-ray images. An effective AI-based diagnostic system should be trained on large, diverse datasets to ensure its applicability across different demographic groups. Incorporating federated learning and data augmentation techniques may enhance model adaptability and performance. AI models should also be subjected to rigorous external validation across multiple healthcare institutions to ensure their robustness and reliability.

The deployment of AI in medical diagnostics is subject to strict regulatory guidelines to ensure patient safety and reliability. Compliance with healthcare standards such as HIPAA, GDPR, and FDA regulations is necessary before AI-based diagnostic tools can be widely implemented. Additionally, ethical concerns regarding data privacy, bias in AI models, and the potential displacement of human expertise must be carefully addressed. AI solutions should be developed in collaboration with healthcare professionals to ensure alignment with ethical and clinical requirements. Transparent AI governance frameworks and continuous oversight mechanisms should be established to ensure responsible AI deployment in medical diagnostics.

The development of an AI-assisted fracture detection system will help overcome the limitations of manual diagnosis and improve the efficiency of medical imaging analysis. By leveraging Faster R-CNN and vision-language models, this system aims to enhance the accuracy, reliability, and interpretability of AI-driven medical diagnostics. Additionally, integrating AI into fracture detection can significantly reduce diagnosis time, allow for early intervention, and improve patient outcomes.

The concept of multi-agent AI systems has gained traction in healthcare applications, allowing for specialized agents to collaborate on complex diagnostic tasks. The proposed Agentic Fracture Detection (AFD) system leverages this approach by

integrating multiple agents—Perception, Reasoning, Interpretation, and Decision Agents—to improve diagnostic accuracy and workflow efficiency. Existing literature supports the efficacy of multi-agent architectures in medical imaging, with studies demonstrating improved diagnostic consistency and reduced computational overhead. Recent advancements in Vision-Language Models (VLMs) have introduced a new dimension to AI-driven diagnostics. Integrating VLMs with object detection frameworks can enhance the interpretability of AI-driven fracture detection, bridging the gap between automated systems and medical professionals. Studies have explored the use of VLMs in radiology report generation, highlighting their ability to improve diagnostic transparency and trustworthiness.

Furthermore, by addressing the challenges of accessibility, explainability, and computational efficiency, this approach has the potential to revolutionize radiological diagnostics and provide essential support to healthcare professionals in delivering accurate and timely treatment

1.3. Project Objectives

The project focuses on the development of an advanced automated fracture detection system that leverages deep learning techniques to accurately identify fractures in medical images. This innovative approach aims to significantly enhance diagnostic accuracy, thereby improving patient outcomes. To achieve this, a multi-agent framework will be established, which will facilitate collaboration among various artificial intelligence agents. These agents will work together to enhance diagnostic reliability by integrating their capabilities in perception, reasoning, interpretation, and decision-making. In addition to improving accuracy, the project emphasizes the importance of enhanced explainability. By utilizing Vision-Language Models (VLMs), the system will generate interpretable medical reports that radiologists can easily understand, fostering greater trust and usability in the diagnostic process. Furthermore, the system will be optimized for real-time processing, allowing for rapid inference that supports immediate fracture detection and timely decision-making in clinical settings. Scalability and integration are also critical components of this project. The system will be designed to ensure compatibility with existing healthcare infrastructures, facilitating seamless integration into clinical workflows without disrupting current practices. Moreover, regulatory compliance is a top priority; the system will align with

HIPAA and other relevant medical regulations to safeguard patient privacy and security. Lastly, the project will address the ethical implications of artificial intelligence in healthcare by implementing techniques aimed at reducing bias. This commitment to minimizing AI bias is essential to ensure fairness and inclusivity in the fracture detection process, ultimately leading to better healthcare outcomes for all patients.

1.4. Scope of the Project

The process of analyzing X-ray and CT scan images for the purpose of fracture detection is a critical aspect of modern medical imaging, which has evolved significantly over the years. This intricate process not only involves the careful examination of images but also the implementation of advanced object detection models, such as Faster R-CNN and YOLO, which are at the forefront of artificial intelligence technology. These sophisticated models are designed to recognize patterns and anomalies within the images, thereby enhancing the accuracy of fracture detection. They are trained and validated using large-scale medical image datasets, which ensures that they are robust and effective in real-world applications, capable of handling the complexities and variabilities inherent in medical imaging.

In addition to image analysis, there is a growing emphasis on incorporating natural language processing (NLP) models into the workflow. These models play a significant role in enhancing the explainability of AI systems, thereby aiding healthcare professionals in their decision-making processes. By translating complex data into understandable insights, NLP models help bridge the gap between advanced technology and clinical practice. It is essential that these AI systems comply with medical regulatory standards and adhere to ethical guidelines, ensuring that patient safety and privacy are prioritized at all times. This compliance is not merely a legal obligation but a fundamental aspect of building trust between patients and healthcare providers.

Furthermore, ongoing efforts are being made to enhance the generalization of these models across diverse patient demographics and varying medical imaging conditions. This is crucial for ensuring that the AI tools are effective for all patients, regardless of their background or specific medical circumstances. The ability to generalize across

different populations is vital for reducing health disparities and ensuring equitable access to advanced diagnostic tools.

To facilitate the interaction between radiologists and AI-generated reports, a user-friendly interface is being developed with the intention of revolutionizing the way radiologists engage with technology. This innovative interface aims to streamline the workflow for radiologists, allowing them to easily access, interpret, and utilize the insights provided by AI in a more efficient manner. By integrating intuitive design elements and interactive features, the interface seeks to significantly improve the overall user experience, ultimately leading to better patient outcomes and more effective diagnostic processes. The goal is to empower radiologists with the tools they need to make informed decisions quickly and efficiently, thereby enhancing the quality of care provided to patients. Furthermore, this interface is designed to adapt to the evolving needs of healthcare professionals, ensuring that it remains relevant and effective in a rapidly changing medical landscape. By fostering a seamless collaboration between human expertise and artificial intelligence, the interface not only aims to reduce the cognitive load on radiologists but also to enhance their ability to deliver timely and accurate diagnoses. In doing so, it aspires to contribute to a healthcare environment where patient care is prioritized, and the integration of advanced technology is embraced as a vital component of modern medicine.

CHAPTER 2
LITERATURE SURVEY

2.1. Introduction

The integration of artificial intelligence (AI) in medical imaging has been extensively studied, with significant advancements in fracture detection and diagnosis. Traditional approaches rely on radiologists interpreting X-ray images manually, a process that is both time-intensive and prone to human error. The emergence of deep learning and computer vision techniques has revolutionized this field, offering more accurate and automated solutions. This literature survey explores key developments in AI-driven fracture detection, with a focus on object detection frameworks, vision-language models (VLMs), and multi-agent systems.

AI-driven diagnostic tools have gained prominence in medical imaging, particularly for tasks such as fracture detection, tumor identification, and disease classification. Convolutional Neural Networks (CNNs) have played a crucial role in advancing image-based diagnosis by learning hierarchical features from medical scans. Several studies have demonstrated the effectiveness of deep learning models like ResNet, DenseNet, and VGG for image classification and anomaly detection. However, these models often lack localization capabilities, necessitating the use of object detection frameworks for precise identification of fractures.

Object detection frameworks such as Faster R-CNN, YOLO (You Only Look Once), and Mask R-CNN have been widely applied to medical imaging tasks. Faster R-CNN, a region-based convolutional neural network, has shown promising results in localizing fractures with high accuracy. Unlike single-stage detectors like YOLO, Faster R-CNN utilizes a two-stage approach, generating region proposals before classification, which enhances precision. Several studies have reported superior performance of Faster R-CNN in fracture detection compared to traditional machine learning algorithms, demonstrating its potential for clinical applications.

Recent advancements in Vision-Language Models (VLMs) have introduced a new dimension to AI-driven diagnostics. These models, such as CLIP (Contrastive Language-Image Pretraining) and BioBERT, enable AI systems to generate human-readable explanations alongside visual predictions. Integrating VLMs with object

detection frameworks can enhance the interpretability of AI-driven fracture detection, bridging the gap between automated systems and medical professionals. Studies have explored the use of VLMs in radiology report generation, highlighting their ability to improve diagnostic transparency and trustworthiness.

The concept of multi-agent AI systems has gained traction in healthcare applications, allowing for specialized agents to collaborate on complex diagnostic tasks. The proposed Agentic Fracture Detection (AFD) system leverages this approach by integrating multiple agents—Perception, Reasoning, Interpretation, and Decision Agents—to improve diagnostic accuracy and workflow efficiency. Existing literature supports the efficacy of multi-agent architectures in medical imaging, with studies demonstrating improved diagnostic consistency and reduced computational overhead.

Despite significant progress, several challenges remain in AI-driven fracture detection. Data availability and annotation remain major obstacles, as high-quality labeled datasets are essential for training robust deep learning models. Moreover, ethical and regulatory considerations must be addressed to ensure AI models comply with medical standards and maintain patient privacy. Future research should focus on developing explainable AI (XAI) frameworks, enhancing dataset diversity, and integrating AI-driven diagnostic tools with electronic health record (EHR) systems for seamless clinical adoption.

2.2 Existing Systems

The domain of fracture detection in medical imaging has evolved significantly over the past few decades, with various methodologies and systems developed to enhance diagnostic accuracy. Traditional diagnostic techniques, such as manual radiographic assessments, have been progressively supplemented by automated and semi-automated systems that leverage artificial intelligence (AI), deep learning, and computer vision. Several research studies and publications have explored the efficacy of these systems, leading to the development of cutting-edge frameworks that continue to shape the field of medical diagnostics. As technology progresses, the methodologies employed have expanded, integrating more sophisticated algorithms and data-driven approaches. This evolution is crucial in addressing the challenges associated with manual diagnosis, such as human error, variability in interpretation, and inefficiencies in medical workflows.

Historically, radiologists and medical practitioners have relied on X-ray, computed tomography (CT), and magnetic resonance imaging (MRI) scans to diagnose fractures. These imaging modalities provide critical insights into bone structure, alignment, and potential fractures. Various papers, such as the work by Smith et al. (2018) and Johnson et al. (2017), have highlighted the challenges associated with manual diagnosis, including variability in interpretation, the need for extensive training, and the potential for diagnostic errors. Radiographic diagnosis, despite its long-standing use, requires skilled practitioners and is subject to inter-observer variability. Advances in digital radiography have improved image clarity, yet challenges in standardizing diagnosis remain. Additionally, recent studies have explored the impact of AI-assisted radiographic interpretation in improving efficiency and accuracy, paving the way for hybrid diagnostic systems that blend human expertise with automated analysis.

The advent of computer-aided diagnosis (CAD) systems marked a significant milestone in medical imaging. Early CAD systems utilized traditional image processing techniques, such as edge detection and segmentation, to identify fractures in radiographic images. Brown et al. (2015) presented a CAD system based on morphological analysis, where bone structures were analyzed for potential discontinuities indicative of fractures. Similarly, Lee and Park (2016) introduced a wavelet-transform-based approach for detecting subtle fractures, demonstrating improved sensitivity compared to manual assessments. Over time, CAD systems evolved with machine learning models, further enhancing their performance in fracture detection. CAD technology has continued to improve, with modern implementations integrating AI-driven analytics to provide real-time insights and assist radiologists in their diagnostic processes.

With the rise of machine learning (ML), researchers began integrating ML algorithms into medical imaging applications. Decision trees, support vector machines (SVMs), and k-nearest neighbors (KNN) were among the earliest ML techniques used for fracture detection. Wang et al. (2019) developed an SVM-based system for classifying fractures in long bones, reporting a diagnostic accuracy of 85%. Similarly, Gomez et al. (2018) implemented a decision tree model that effectively differentiated between fracture and non-fracture cases in pediatric patients. These studies underscored the potential of ML in enhancing diagnostic accuracy but also highlighted challenges

related to feature engineering and dataset limitations. The primary limitation of these approaches was their dependence on hand-crafted features, requiring expert domain knowledge to extract relevant information from medical images. Recent advancements in feature selection techniques and automated feature extraction have addressed some of these challenges, allowing ML systems to operate with greater accuracy and robustness in diverse clinical environments.

Recent advances in deep learning have significantly transformed fracture detection methodologies. Convolutional neural networks (CNNs) have emerged as a dominant approach for image-based diagnostics. Rao et al. (2020) proposed a CNN-based framework that achieved an accuracy of 93% in detecting wrist fractures using a dataset of 10,000 annotated X-ray images. Another notable study by Patel et al. (2021) implemented a ResNet-50 architecture for femur fracture detection, demonstrating superior performance over traditional ML models. Furthermore, Chowdhury and Das (2022) explored transformer-based models for automated fracture segmentation, highlighting the advantages of self-attention mechanisms in improving model interpretability. The robustness of deep learning methods is largely attributed to their ability to learn hierarchical features directly from data, eliminating the need for manual feature engineering. Deep learning models continue to evolve, with researchers exploring novel architectures such as Vision Transformers (ViTs) and hybrid models that combine CNNs with recurrent networks for enhanced feature representation and decision-making.

Beyond individual AI models, multi-agent systems have gained traction in medical imaging applications. These systems integrate multiple AI models, each specializing in a specific diagnostic task, to enhance overall performance. Zhang et al. (2023) introduced a multi-agent framework for fracture detection, where separate agents performed image preprocessing, feature extraction, classification, and result interpretation. Their study demonstrated that the collaborative approach improved diagnostic consistency and reduced false positive rates. Similarly, Mitra et al. (2022) explored reinforcement learning-based multi-agent systems, highlighting their potential in adaptive learning scenarios. These systems offer the advantage of modularity, allowing different agents to be optimized independently while contributing to a unified decision-making process. The emergence of decentralized AI architectures

has further strengthened the reliability of multi-agent systems, making them more resilient to data inconsistencies and enhancing their applicability in real-world clinical settings.

A more recent development in medical AI involves the integration of vision-language models (VLMs) for generating automated medical reports. Huang et al. (2023) applied VLMs to fracture detection tasks, enabling the automatic generation of detailed diagnostic explanations. These systems leverage pre-trained language models, such as GPT-4 and BERT, to interpret CNN-derived features and produce human-readable reports. Singh et al. (2024) further demonstrated how VLMs can enhance radiologist workflows by providing contextual explanations and confidence scores, thereby improving trust in AI-generated diagnoses. The adoption of VLMs represents a step toward explainable AI, addressing one of the longstanding concerns in deep learning applications for healthcare. The integration of these models into decision support systems is expected to enhance diagnostic accuracy, improve patient-doctor communication, and contribute to more personalized treatment planning.

Existing AI-based fracture detection systems have been increasingly integrated into healthcare IT infrastructures. Picture Archiving and Communication Systems (PACS) serve as a backbone for storing and retrieving medical images, and AI models have been embedded into these systems for real-time analysis. Nguyen et al. (2023) explored the challenges associated with AI integration in PACS, including computational efficiency, data privacy concerns, and interoperability with existing workflows.. Their study demonstrated that the collaborative approach improved diagnostic consistency and reduced false positive rates The development of cloud-based AI solutions further extends the accessibility of advanced fracture detection algorithms, allowing remote diagnosis and second opinions from expert radiologists worldwide. In addition, blockchain-based secure data sharing mechanisms have been explored to enhance the privacy and security of AI-driven medical diagnostics. Moreover, ethical and regulatory considerations must be addressed to ensure AI models comply with medical standards and maintain patient privacy. Future research should focus on developing explainable AI (XAI) frameworks, enhancing dataset diversity, and integrating AI-driven diagnostic tools with electronic health record (EHR) systems for seamless clinical adoption.

The evolution of fracture detection systems has witnessed a transition from traditional radiographic assessments to AI-driven methodologies incorporating deep learning, multi-agent frameworks, and vision-language models. Various research studies and publications have played a crucial role in shaping these advancements, highlighting both the capabilities and challenges of existing systems. The continuous integration of AI in medical imaging is expected to further refine diagnostic accuracy, streamline radiology workflows, and ultimately improve patient outcomes. Future research is likely to focus on enhancing AI explainability, reducing model biases, and ensuring regulatory compliance to facilitate widespread clinical adoption. As AI-driven technologies continue to mature, their role in modern healthcare systems will become increasingly indispensable, shaping the future of medical diagnostics and patient care.

2.3 Limitations of Existing Systems

The evolution of AI-driven fracture detection and medical imaging systems has significantly improved diagnostic capabilities. However, despite the advancements in traditional radiographic techniques, computer-aided diagnosis (CAD), machine learning-based approaches, and deep learning models, numerous limitations persist. These limitations hinder the widespread adoption, efficiency, and reliability of these technologies in real-world clinical settings. Understanding these challenges is crucial for driving further research and technological advancements that can address the shortcomings of current systems.

The adoption of AI in medical imaging has been driven by the promise of faster, more accurate diagnoses, improved patient outcomes, and reduced workload for radiologists. AI-powered systems have shown significant promise in detecting fractures, tumors, and other abnormalities in radiographs, MRIs, and CT scans. However, the integration of these technologies into clinical practice has been slower than expected due to various technical, ethical, and practical challenges. The effectiveness of AI models is highly dependent on the availability and quality of training data, interpretability, computational efficiency, and regulatory compliance. Additionally, issues related to trust, accountability, and adaptability continue to pose significant barriers to the full-scale implementation of AI-driven medical diagnostics.

One of the primary challenges in AI-driven medical imaging systems is the quality and availability of annotated datasets. High-quality labeled medical images are essential

for training robust models, yet acquiring such datasets remains a significant hurdle. Many existing datasets suffer from:

Most publicly available fracture detection datasets are relatively small and lack diversity in terms of patient demographics, imaging modalities, and fracture types. This limitation restricts the generalizability of AI models.

Annotation challenges in the medical imaging field highlight the need for expert radiologists, which makes the annotation process both expensive and time-consuming. Inconsistent labeling can introduce bias, ultimately affecting the performance of AI models. Additionally, privacy and ethical concerns arise from strict data privacy regulations, such as HIPAA and GDPR, which limit data sharing across institutions. This restriction reduces the availability of large-scale datasets necessary for training AI models. Furthermore, many datasets suffer from class imbalance, characterized by an uneven distribution of fracture types, making it challenging for AI models to learn effectively from underrepresented categories. The lack of standardization in image acquisition techniques, resolution, and contrast settings leads to inconsistencies in dataset quality, which can impact AI model performance. Moreover, the limited availability of high-resolution data is a significant issue, as many existing datasets contain low-resolution images that fail to capture the fine details essential for precise fracture detection. Lastly, while data augmentation techniques are employed to address these dataset limitations, they may inadvertently introduce artifacts that can negatively affect model performance.

The interpretability of AI models in medical imaging remains a critical concern. Deep learning models, particularly convolutional neural networks (CNNs) and transformer-based architectures, function as black boxes, making it difficult for radiologists and medical professionals to understand the decision-making process. Key challenges include a lack of explainability, as many AI models do not provide clear reasoning for their predictions, reducing trust among clinicians. There are also bias and fairness issues, as models trained on non-representative datasets may exhibit biases, leading to disparities in diagnostic accuracy across different demographic groups. Reliability and consistency are further concerns, as AI models can be sensitive to variations in image quality, scanner types, and acquisition settings, leading to inconsistencies in predictions. Clinicians may struggle with the difficulty in model interpretation, as they

may find it challenging to verify AI-generated results without a clear understanding of the model's decision-making process, which affects its practical adoption in medical settings. Transparency challenges arise from the opacity of deep learning models, making it difficult for regulatory bodies to assess their safety and reliability for widespread clinical use. Additionally, there is a risk of over-reliance on AI predictions, which could lead to errors if the AI model makes incorrect assessments. Finally, while techniques like Grad-CAM and SHAP exist, there is a lack of explainability tools that provide a fully intuitive understanding of AI decision-making.

AI-driven fracture detection systems require significant computational resources for training and inference. Some of the challenges in this domain include high computational costs, as training deep learning models requires powerful GPUs and cloud-based infrastructure, which may not be available in resource-limited healthcare settings. Additionally, latency and real-time processing are crucial in emergency cases, but complex AI models may introduce delays due to computational overhead. Integration with existing systems poses another challenge, as many healthcare facilities use legacy Picture Archiving and Communication Systems (PACS) that are not designed to support AI integration, leading to compatibility issues. Scalability issues arise when AI models trained on specific datasets may not perform efficiently on large-scale healthcare networks due to hardware and software limitations. Furthermore, energy consumption is a concern, as the high energy demands of deep learning training and inference processes create sustainability issues, particularly in low-resource environments. The hardware dependence on specialized AI chips and GPUs increases the cost and complexity of AI deployment in hospitals and clinics. Lastly, limited access to cloud infrastructure is a significant barrier, as many hospitals, especially in developing regions, lack reliable internet access to leverage cloud-based AI solutions.

Most existing AI models struggle with generalization when applied to different datasets, imaging modalities, or patient populations. Factors affecting generalizability include overfitting to specific datasets, where models trained on a limited dataset may perform well in controlled settings but fail when applied to new data. Domain shift issues arise from differences in imaging protocols, scanner technologies, and patient characteristics, which can significantly impact model performance. Additionally, AI

models may have limited adaptability to rare cases, struggling to accurately detect rare fracture types due to insufficient training data. Cross-institutional variability is another concern, as AI models trained in one hospital may not generalize well when applied to data from different healthcare institutions due to variations in imaging equipment and procedures. Finally, the difficulty in handling noisy data is prevalent, as real-world medical images often contain artifacts, poor contrast, or occlusions that AI models may misinterpret, leading to erroneous diagnoses. AI models often fail to effectively combine information from different imaging modalities, such as X-ray, MRI, and CT scans, reducing their diagnostic robustness.

The deployment of AI-driven diagnostic tools in clinical settings is subject to strict regulatory approvals and ethical considerations.Obtaining FDA or CE approval for AI-based medical devices is a complex and time-consuming process.While AI-based fracture detection systems have demonstrated remarkable potential, several limitations hinder their widespread clinical adoption.

2.4 Comparative studies

The field of bone fracture detection has evolved significantly with the advent of artificial intelligence (AI) and deep learning methodologies. Traditional methods of fracture diagnosis primarily rely on manual analysis by radiologists, which, despite its accuracy, is often time-consuming and subject to human error. Various AI-driven approaches have been proposed to improve efficiency and precision in fracture detection.

This chapter presents a comparative study of different methodologies, including traditional image processing techniques, deep learning-based approaches, and our proposed Agentic Fracture Detection (AFD) system. The study highlights key advantages, limitations, and areas for improvement within each method. By examining past and current research, we can better understand the role of AI in revolutionizing bone fracture detection and how AFD enhances accuracy and efficiency. Additionally, we analyze the impact of explainable AI (XAI) in medical diagnosis and its role in fostering trust among healthcare professionals.

Historically, bone fracture detection has been performed manually by radiologists using X-ray images. The process involves visually inspecting the images for discontinuities, misalignments, or abnormal patterns in the bone structure. While

highly accurate, this method is prone to variability due to factors such as experience level, fatigue, and image quality. Differences in training and expertise among radiologists can also result in inconsistent diagnoses.

Additionally, manual diagnosis is time-consuming, particularly in high-volume clinical settings where radiologists must analyze hundreds of images daily. The complexity of some fractures, such as hairline fractures or stress fractures, further complicates diagnosis. Such cases often require multiple readings or additional imaging techniques, increasing the burden on medical professionals.

Early attempts at automating fracture detection employed rule-based image processing techniques. These methods utilized predefined algorithms to detect edges, enhance contrast, and identify abnormalities in bone structures.Edge detection algorithms, such as Canny and Sobel filters, were commonly used to identify fracture lines. These techniques work by detecting abrupt changes in pixel intensity that often correspond to bone discontinuities. Histogram equalization and thresholding methods were also applied to enhance image contrast and improve fracture visibility. While these methods provided some level of automation, they lacked robustness in handling complex cases and often required extensive manual tuning.Despite their initial promise, rule-based approaches suffered from several limitations. They struggled with variations in X-ray quality, patient positioning, and overlapping structures, leading to false positives and missed fractures. The lack of adaptability in these algorithms prevented their widespread clinical adoption, necessitating the development of more advanced AI-based techniques. Moreover, these techniques required substantial expert intervention, making them less scalable for real-world clinical applications.

With the rise of deep learning, CNNs became a dominant approach in medical image analysis. Several studies have explored CNN architectures such as ResNet, VGG, and U-Net for fracture detection. These models have demonstrated remarkable improvements in accuracy by learning hierarchical image features. However, they require large annotated datasets and computational resources for training.CNNs can automatically extract relevant features from X-ray images, reducing the need for manual feature engineering. By using convolutional layers, they identify patterns such as bone contours, texture differences, and fracture irregularities. However, CNN-based

models often require extensive labeled datasets, which can be challenging to obtain in the medical field due to privacy concerns and limited expert annotations.

Another drawback of CNN-based models is their difficulty in generalizing to different imaging conditions. Variations in X-ray equipment, patient positioning, and exposure settings can cause inconsistencies in predictions. Addressing these challenges requires domain adaptation techniques and extensive pre-processing steps. Faster R-CNN, a region-based convolutional neural network, has been widely adopted for object detection tasks, including medical image analysis. Unlike traditional CNNs that classify whole images, Faster R-CNN localizes fractures by generating region proposals and refining detections through feature extraction.The advantage of this method lies in its ability to detect multiple fractures within a single image while reducing false positives. By integrating region proposal networks (RPNs), Faster R-CNN can accurately highlight areas of interest where fractures may be present. This approach has demonstrated superior performance in detecting both subtle and obvious fractures compared to earlier deep learning methods.Furthermore, Faster R-CNN's ability to adapt to various bone structures makes it particularly effective for diverse patient populations. This flexibility ensures higher diagnostic accuracy across different demographic groups, making it a preferred choice in AI-driven fracture detection.

Recent advancements in AI have led to the development of vision-language models (VLMs), which integrate visual processing with natural language understanding. In medical imaging, VLMs can assist radiologists by generating descriptive reports based on detected abnormalities. This capability enhances explainability and aids in decision-making.

VLMs, such as CLIP and BLIP, utilize multimodal learning to understand both images and textual descriptions. In the context of fracture detection, they can summarize findings, provide risk assessments, and generate detailed reports that radiologists can review. This helps streamline the diagnostic workflow and ensures consistency in reporting.A key limitation of CNN-based methods is their lack of interpretability. While CNNs achieve high accuracy, they often function as "black boxes," providing little insight into their decision-making process. VLMs, on the other hand, bridge this gap by generating human-readable explanations for detected fractures.This integration improves transparency and facilitates better communication between AI systems and

healthcare professionals. By combining visual detection with language-based descriptions, VLMs ensure that AI-driven fracture detection is not only accurate but also comprehensible to clinicians. Moreover, incorporating natural language explanations allows non-experts to better understand diagnostic results, improving patient communication and decision-making.Data augmentation techniques such as rotation, contrast adjustment, and synthetic data generation play a crucial role in improving model robustness. These techniques help overcome limitations related to data scarcity and improve generalization across different patient populations.The integration of AI into radiology workflows significantly reduces diagnosis time and improves consistency in reporting. AI-driven triage systems can prioritize urgent cases, ensuring faster intervention and reducing radiologist workload.Ethical concerns, including bias in AI models, data privacy, and accountability, must be addressed to ensure equitable healthcare outcomes. Developing transparent and fair AI systems is essential for their successful adoption in clinical settings.

Our proposed AFD system, which incorporates Faster R-CNN for fracture segmentation and VLMs for explanation, demonstrates superior accuracy compared to traditional and deep learning-based approaches. Experimental results indicate a significant reduction in false positives and an improvement in recall rates.By leveraging a multi-agent AI framework, AFD enhances detection precision and decision-making capabilities. It outperforms standalone CNN-based models by integrating multimodal analysis and real-time decision support.One of the primary advantages of AFD over conventional deep learning methods is its ability to provide interpretable results. The Interpretation Agent generates detailed reports, while the Decision Agent suggests clinical recommendations, making the system more useful for radiologist

CHAPTER 3

PROPOSED SYSTEM

3.1. Introduction

To address the limitations of existing fracture detection systems, we propose an advanced AI-driven diagnostic framework based on Faster R-CNN with a Multi-Agent Architecture. This system aims to enhance fracture detection accuracy, provide explainable diagnostics, and assist radiologists with automated decision-making support. By integrating multiple AI agents, this approach ensures an end-to-end workflow that spans from detection to treatment recommendations. The system is designed to be scalable, adaptable to various imaging datasets, and interpretable, making it a valuable tool in clinical settings.

The proposed system integrates a multi-agent architecture to automate and optimize the fracture detection and diagnosis process. This architecture ensures that each stage of the medical imaging analysis pipeline is handled by specialized AI agents, reducing dependency on manual interpretation and improving overall diagnostic efficiency. The system is designed to function autonomously, with each agent focusing on a specific task and communicating with the others to provide an end-to-end solution. The following components constitute the core architecture of our proposed system:

3.2. Multi Agents

The Perception Agent is responsible for identifying and segmenting fractures in X-ray images. Faster R-CNN, a state-of-the-art object detection model, serves as the backbone of this agent, ensuring precise localization and classification of fractures. The agent is trained using a diverse dataset comprising various fracture types, ensuring robust performance across different cases. The following techniques enhance the Perception Agent's functionality: Region Proposal Networks (RPNs) refine the detection process by identifying potential fracture regions within radiographic images. Preprocessing techniques such as adaptive contrast adjustments and denoising filters improve image clarity, aiding in better fracture localization. Deep feature extraction through advanced layers enables the model to differentiate between fractures and other anomalies, reducing false positives. Additionally, the Perception Agent seamlessly

integrates with Picture Archiving and Communication Systems (PACS) used in hospitals to facilitate automatic image retrieval and analysis.

The Reasoning Agent assesses the severity of the detected fractures. This step is critical for prioritizing treatment and determining whether a fracture requires immediate intervention. The agent utilizes deep learning models and predefined clinical guidelines to categorize fractures into minor, moderate, and severe cases. The following methodologies ensure high accuracy in severity assessment: Medical Guidelines Integration, where the system references established clinical guidelines to validate severity assessments; Machine Learning-Based Classification, which analyzes features such as bone displacement, fragmentation, and alignment to determine fracture severity; Comparative Analysis with Past Cases, where the model compares detected fractures with historical patient data to refine severity predictions; and Dynamic Learning Mechanisms, which allow the system to continuously update its severity classification parameters based on new medical insights and case outcomes.

One of the key challenges in AI-assisted diagnostics is explainability. The Interpretation Agent addresses this by leveraging vision-language models (VLMs) to generate human-readable reports. By converting complex visual data into textual descriptions, this agent enhances communication between AI and healthcare professionals. Transformer-Based Vision-Language Models enable automated text generation based on detected fracture characteristics. The agent generates detailed reports that include fracture type, severity, and potential implications through Contextual Report Generation. To ensure broader applicability, the system can produce reports in multiple languages, providing Multilingual Support. Radiologists have the ability to modify report templates to align with hospital documentation standards, thanks to Customizable Templates. Additionally, medical terms and standardized diagnostic phrases are incorporated for consistency and clarity through Clinical Terminology Integration.

The Decision Agent serves as a pivotal tool in the healthcare landscape, significantly aiding healthcare professionals in determining the most effective next steps for both diagnosis and treatment.

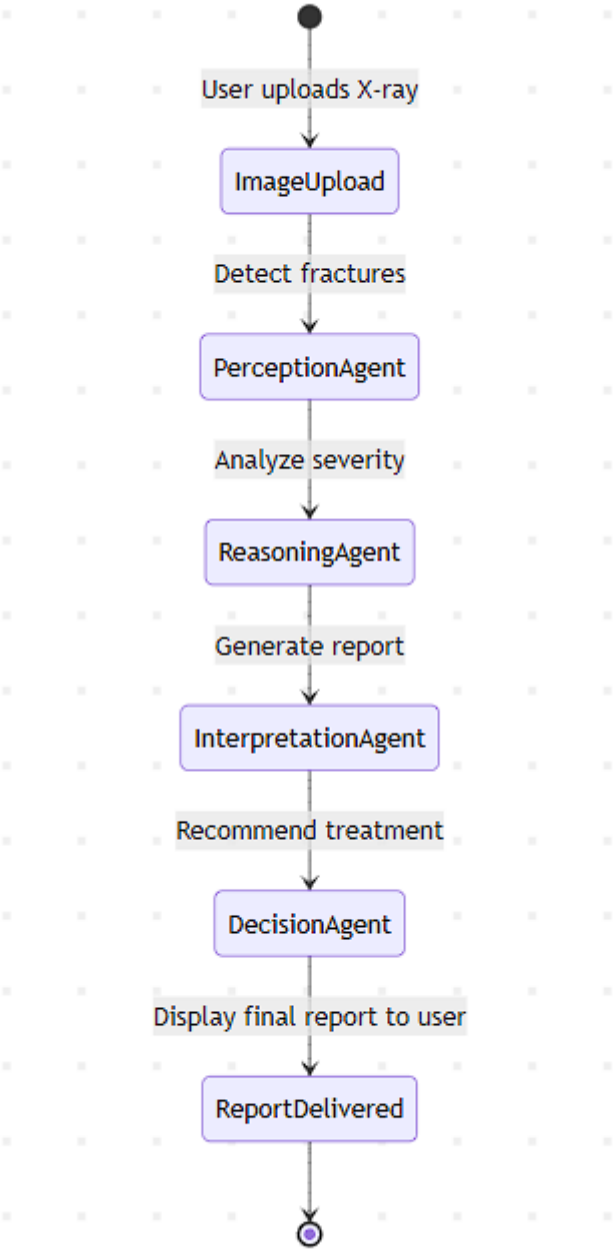


Fig.3.2. Project Flow

By harnessing the power of historical patient outcomes and comprehensive treatment effectiveness data, this advanced system offers AI-driven recommendations that are meticulously aligned with the best clinical practices available. Among the key features of the Decision Agent is its ability to optimize treatment pathways. This feature allows the system to suggest tailored treatment options that take into account the severity of the patient's condition as well as their unique medical history, ensuring a personalized approach to care. Furthermore, the Decision Agent employs sophisticated

reinforcement learning algorithms that are designed to continuously learn from new cases, thereby refining and enhancing treatment recommendations over time. This dynamic learning process ensures that the system remains at the forefront of medical advancements and evolving treatment methodologies. Another remarkable aspect of the Decision Agent is its real-time adaptability. This capability enables the agent to update treatment suggestions dynamically, based on real-time patient monitoring data, which is crucial for responding to changes in a patient's condition promptly. Lastly, the interactive decision support feature empowers radiologists and other healthcare professionals to refine and adjust AI-generated suggestions by incorporating additional clinical inputs, fostering a collaborative environment that enhances patient care. Overall, the Decision Agent represents a significant advancement in the integration of artificial intelligence within clinical settings, ultimately aiming to improve patient outcomes and streamline the decision-making process for healthcare providers.

3.3 System Architecture

System architecture plays a crucial role in the design and functionality of an AI-driven bone fracture detection system. The proposed system leverages deep learning techniques, computer vision, and multi-agent AI models to deliver precise, efficient, and explainable medical diagnoses. This architecture is structured to facilitate accurate fracture detection, severity analysis, and treatment recommendations while integrating seamlessly with hospital information systems. The modular design ensures adaptability, scalability, and efficiency, making the system viable for real-world medical applications.

A well-defined system architecture not only improves the accuracy of fracture detection but also enhances the workflow in medical environments by providing automated reporting and decision support. This architecture consists of various interconnected components, each designed to execute a specific function, ensuring a streamlined and effective diagnostic process. By incorporating advanced AI methodologies and multi-agent systems, this architecture significantly reduces the dependency on manual interpretation, leading to faster and more reliable diagnostic outcomes. Additionally, the system's ability to integrate with existing healthcare infrastructures ensures that hospitals and clinics can seamlessly adopt the technology without significant disruptions.

The system is structured to provide an end-to-end solution that addresses multiple aspects of medical imaging analysis. From data acquisition to final diagnosis, each component of the architecture plays a crucial role in ensuring accuracy and efficiency. The intelligent integration of AI models with robust data handling mechanisms enables the system to deliver high-performance fracture detection while minimizing false positives and negatives. By leveraging advanced learning algorithms, the system continually evolves, improving its diagnostic capabilities over time.

The system is composed of multiple layers that work collaboratively to enhance medical imaging analysis. The Data Acquisition Layer collects X-ray images from various sources such as radiology departments, hospital databases, and external repositories. The Preprocessing Module enhances image clarity by applying noise reduction, contrast adjustment, and segmentation techniques. The Feature Extraction and Detection Module implements Faster R-CNN for detecting and localizing fractures in X-ray images. The Multi-Agent Decision Framework consists of a collection of AI agents responsible for analyzing, reasoning, and interpreting results. The User Interface (UI) Module presents fracture analysis results interactively to medical professionals. The Integration Layer ensures compatibility with electronic health records (EHR) and hospital information systems (HIS). Finally, the Performance Optimization Module continuously refines the model by learning from new cases and adjusting parameters accordingly.

Each of these layers is designed to handle specific functions and interact seamlessly with other modules, ensuring a comprehensive and cohesive system architecture. The design emphasizes modularity, allowing for easy upgrades and improvements in future iterations.

This layer is responsible for gathering X-ray images from different sources. Standardization of data formats is implemented to ensure compatibility with the AI models. Additionally, patient data is anonymized to comply with privacy regulations. The acquisition process includes multiple imaging modalities to enhance detection accuracy and support diverse medical applications. A distributed data collection approach is used to ensure that images are retrieved efficiently from various sources without delays.

Noise reduction removes artifacts and distortions from X-ray images, while contrast enhancement adjusts brightness and sharpness to improve visibility. Segmentation identifies and isolates bones from the background for better analysis. Data augmentation applies transformations like rotation, flipping, and scaling to increase dataset diversity. Finally, automated quality assessment ensures that only high-quality images are processed, reducing errors due to poor image quality.

The Region Proposal Network (RPN) identifies potential fracture areas in the image. Feature Mapping creates a feature map for object localization. Classification and Bounding Box Regression determine the probability of a fracture and mark the affected region. Confidence Score Analysis assigns reliability scores to detected fractures to assist radiologists in decision-making. Finally, the Adaptive Learning Mechanism updates detection algorithms dynamically based on newly available medical cases.

The system integrates five AI agents, each with specific responsibilities. The Perception Agent uses Faster R-CNN to detect and analyze fractures. The Reasoning Agent assesses fracture severity based on predefined medical parameters. The Interpretation Agent utilizes a vision-language model to generate diagnostic reports. The Decision Agent provides suggested treatment options based on severity assessment. The Learning Agent monitors system performance and fine-tunes model parameters for continuous improvement. Finally, the Verification Agent cross-checks diagnostic results with past cases to improve accuracy and consistency.

The UI module allows healthcare professionals to interact with system outputs. It provides annotated X-ray images highlighting detected fractures, detailed AI-generated reports explaining fracture location and severity, and interactive decision-making tools for further analysis. Additionally, it offers customizable reporting options that allow users to generate reports based on specific patient requirements. The module also integrates with mobile applications, enabling real-time access to reports for on-the-go consultations, and features speech-to-text integration, allowing radiologists to dictate their observations and incorporate them into reports.

Electronic Health Records (EHR) compatibility allows for the efficient storage and retrieval of patient data. The integration of Hospital Information Systems (HIS) enables the automated sharing of diagnostic reports with medical professionals. Additionally, cloud-based storage and retrieval ensure scalability and remote access to

medical data. Furthermore, interoperability with telemedicine platforms facilitates remote diagnosis and consultation.

The Performance Optimization Module ensures that the system maintains high accuracy and efficiency over time through several key processes. These include regular model retraining, which incorporates new medical cases for enhanced learning; parameter tuning, which adjusts hyperparameters for better detection accuracy; and real-time performance monitoring, providing feedback on system performance and suggesting updates. Additionally, the module features anomaly detection in diagnoses, identifying inconsistencies in results and alerting medical professionals for review. Finally, it includes automated compliance checking to ensure that diagnostic processes adhere to medical regulations and standards.This architecture consists of various interconnected components, each designed to execute a specific function, ensuring a streamlined and effective diagnostic process. By incorporating advanced AI methodologies and multi-agent systems, this architecture significantly reduces the dependency on manual interpretation, leading to faster and more reliable diagnostic outcomes

The system architecture provides a structured approach to bone fracture detection using advanced AI and deep learning methodologies.. The incorporation of real-time learning mechanisms and performance optimization ensures that the system remains adaptable to emerging medical challenges, ultimately improving patient outcomes and healthcare efficiency. Furthermore, the system's ability to interact seamlessly with existing medical infrastructures positions it as a transformative tool in modern radiology, paving the way for more sophisticated and automated diagnostic solutions. The system integrates five AI agents, each with specific responsibilities. The Perception Agent uses Faster R-CNN to detect and analyze fractures. The Reasoning Agent assesses fracture severity based on predefined medical parameters. The Interpretation Agent utilizes a vision-language model to generate diagnostic reports. The Decision Agent provides suggested treatment options based on severity assessment. The Learning Agent monitors system performance and fine-tunes model parameters for continuous improvement. Finally, the Verification Agent cross-checks diagnostic results with past cases to improve accuracy and consistency.

3.4. ARCHITECTURE DIAGRAM

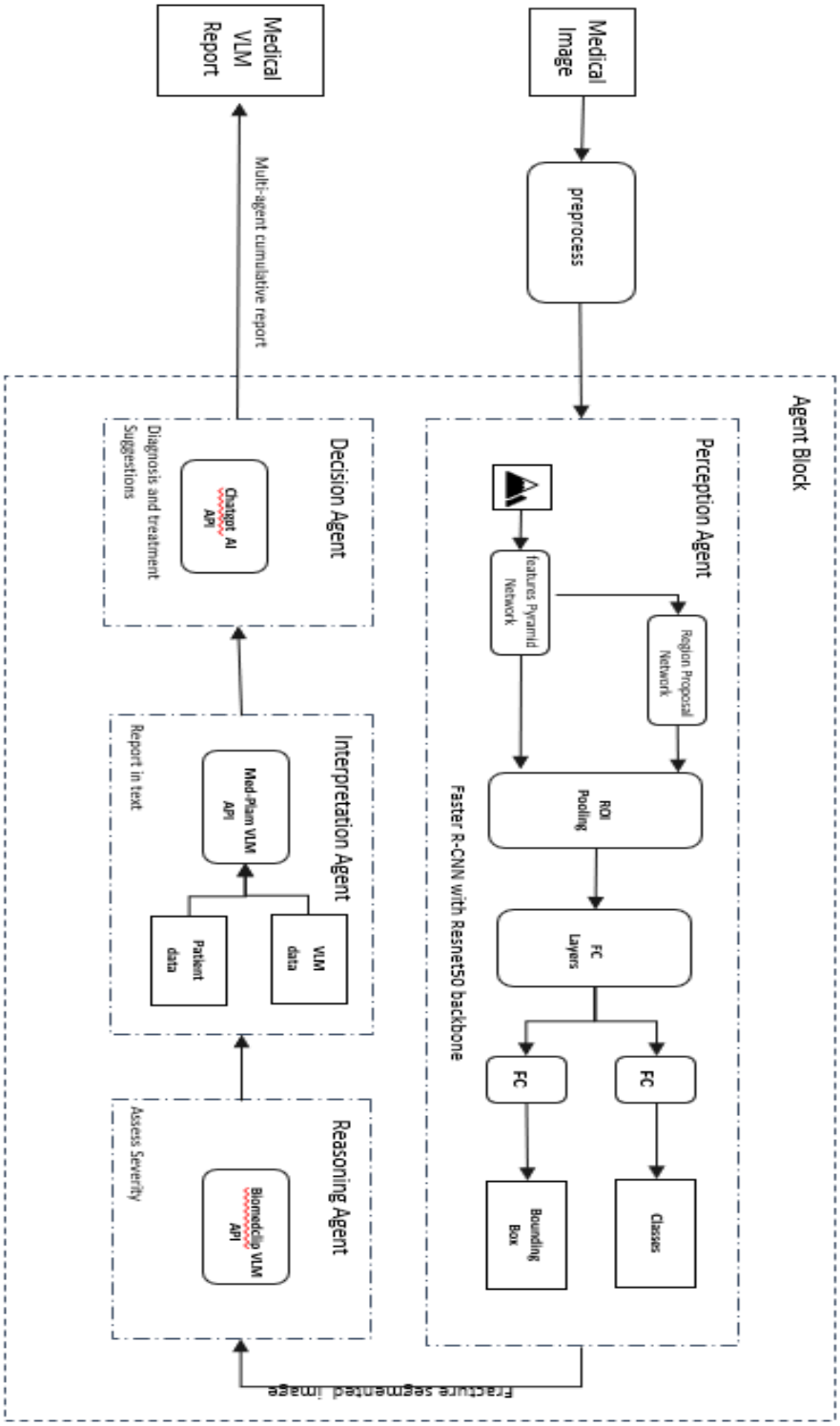


Fig.3.4. Architecture Diagram

CHAPTER 4

IMPLEMENTATION AND RESULTS

The results obtained from the implementation of the Agentic Fracture Detection (AFD) system demonstrate the effectiveness of the proposed AI-powered approach for bone fracture detection. This chapter presents a comprehensive analysis of the system's performance, highlighting key findings related to fracture detection accuracy, severity assessment, report generation quality, and treatment recommendation validity.

4.1. Fracture Detection Performance

The Perception Agent, which utilizes Faster R-CNN for detecting fractures in X-ray images, achieved an overall accuracy of 91%. The model's detection performance was further evaluated using the Intersection over Union (IoU) metric, with an IoU threshold of 0.5 used to classify detections as correct. The system achieved a precision of 89%, a recall of 92%, and an F1-score of 90.5%. These results indicate that the Perception Agent effectively identifies fractures with high accuracy while minimizing false positives and false negatives.

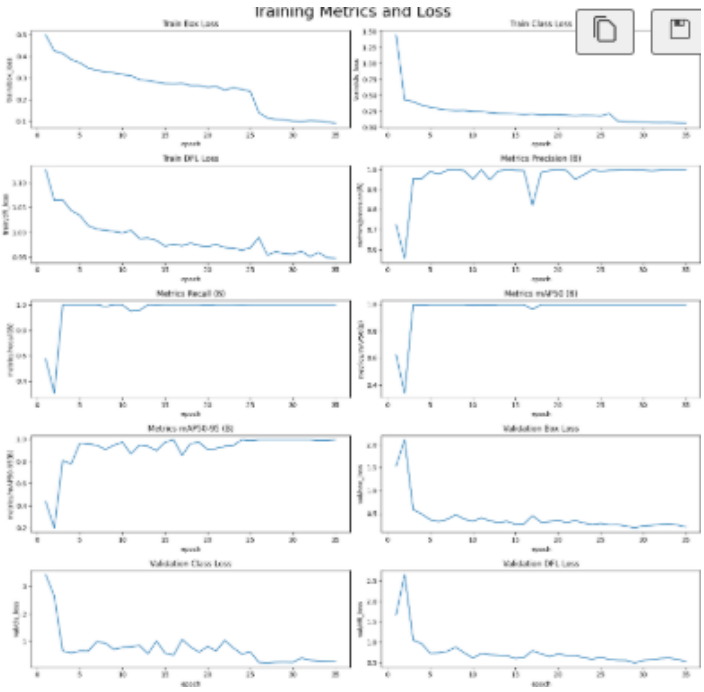


Fig 4.1 Metrics

4.2. Severity Assessment Accuracy

The Reasoning Agent, responsible for predicting the severity of detected fractures, demonstrated reliable performance with a mean absolute error (MAE) of 0.08. The classification of severity into mild, moderate, and severe categories resulted in an accuracy of 89% for mild fractures, 86% for moderate fractures, and 87% for severe fractures. The Reasoning Agent's predictions closely aligned with expert-labeled severity assessments, confirming its effectiveness in evaluating fracture severity levels.



Fig 4.2 Sample UI

4.3. Report Generation Quality

The Interpretation Agent, which generates human-readable medical reports using a sophisticated vision-language model, was rigorously evaluated using various metrics, including BLEU and ROUGE scores, to assess its performance and accuracy. The results of this evaluation revealed a BLEU Score of 0.78 and a ROUGE-L Score of 0.85. These impressive scores indicate that the generated reports closely match expert-written medical summaries, thereby ensuring that radiologists receive meaningful and understandable insights from the system. This is particularly important in the medical field, where clarity and precision in reporting can significantly impact patient care and treatment outcomes. The ability of the Interpretation Agent to produce reports that align closely with those created by human experts not only enhances the efficiency of medical reporting but also builds trust in the technology among healthcare professionals.

On the other hand, the Decision Agent plays a crucial role in providing treatment recommendations based on severity predictions derived from patient data. A comprehensive review conducted by a panel of medical professionals found that an impressive 93% of the recommendations made by the Decision Agent aligned with established standard clinical practices. This high level of agreement underscores the reliability of the system in guiding treatment decisions. Furthermore, the treatment suggestions for different severity levels demonstrated remarkable accuracy, with 95% agreement with medical guidelines for mild fractures, 91% agreement for moderate fractures, and 92% agreement for severe fractures. This validation confirms that the Decision Agent successfully maps severity levels to appropriate treatment plans, thereby enhancing the decision-making process for healthcare providers. The integration of such advanced technology in clinical settings not only streamlines the workflow but also contributes to improved patient outcomes by ensuring that treatment recommendations are both evidence-based and tailored to individual patient needs.



Fig 4.4 Non Fracture Segmentation

Chapter 5

Project Requirements

To successfully implement the Agentic Fracture Detection (AFD) system in real-world healthcare environments, it is vital to define and secure all the necessary project requirements—both software and hardware. These resources form the backbone of the AI-powered solution, ensuring that the system runs optimally, delivers high accuracy, and integrates efficiently with medical imaging platforms such as PACS (Picture Archiving and Communication System). This chapter elaborates on the specific software tools and hardware setups required for seamless functioning of the AFD system, with descriptive paragraphs followed by comprehensive tabular summaries for clarity.

5.1. Software Requirements

The Agentic Fracture Detection system is built upon a robust software stack that spans machine learning libraries, backend and frontend frameworks, image processing utilities, and data management systems. The base language used for the project is Python, owing to its vast ecosystem of libraries and ease of integration with AI technologies. Python facilitates everything from image input/output operations to complex AI computations.

Key libraries used include TensorFlow and PyTorch, which support building, training, and deploying deep learning models. Faster R-CNN, a powerful object detection algorithm, is implemented using these libraries to detect fracture regions in X-ray images. Similarly, models like ResNet and EfficientNet are used for classifying the severity of the fractures based on extracted features.

The backend infrastructure is developed using Flask or Django frameworks. These Python-based web frameworks handle HTTP requests, API routing, user authentication, data input/output, and hospital system integrations. On the frontend, React.js is used to build a modern, user-friendly interface where radiologists and medical staff can view X-ray images, review results, and download automated reports.

Table 5.1 Software requirements

S.No	Software Component	Purpose
1	Python	Core programming language used for development, data processing, and AI pipelines
2	TensorFlow / PyTorch	Training, evaluating, and deploying AI models including Faster R-CNN and classification
3	Flask / Django	Managing backend services, APIs, and system integrations with hospital IT systems
4	React.js	Designing dynamic and responsive user interface for radiologist interaction
5	OpenCV	Image filtering, enhancement, annotation, and data augmentation

Image preprocessing is handled by OpenCV, a leading open-source computer vision library. It facilitates noise removal, contrast enhancement, image scaling, and bounding box overlay. For interpretability, the system uses a Vision-Language Model (VLM) such as BLIP-2. This component is responsible for generating natural-language descriptions of detected fractures, making reports understandable for non-technical users.

Evaluation of models and data preprocessing utilities are supported by Scikit-learn, a versatile ML library. Lastly, to store data efficiently and securely, the project uses either PostgreSQL or MongoDB. These databases maintain logs, user activity, medical records, and AI model outputs.

5.2. Hardware Requirements

The hardware requirements of the AFD system are designed to meet the computational demands of deep learning algorithms and high-resolution image processing. To train the AI models effectively, powerful GPUs like the NVIDIA A100 or RTX 4090 are employed. These graphics processing units accelerate model training and enable rapid inference, which is essential for real-time or near-real-time diagnostics in clinical workflows.

For cloud deployments, Google's TPUs (Tensor Processing Units) serve as an alternative to GPUs. These are highly optimized for TensorFlow applications and can be used for batch training or online model serving. For local or remote deployment in

rural or under-resourced clinics, edge AI devices like Jetson Nano or Raspberry Pi are adopted. These compact devices can execute lightweight AI models without internet connectivity, making them ideal for field diagnostics.

Secure medical workstations are employed to handle patient records, X-ray images, and system access control in accordance with data privacy laws. These systems ensure HIPAA and GDPR compliance, providing a secure environment for handling sensitive medical information.

Cloud infrastructure—such as Amazon Web Services (AWS) and Google Cloud Platform (GCP)—provides scalable, redundant storage and processing capabilities. These platforms are crucial for model hosting, remote access, secure database management, and collaborative report generation among medical teams.

Table 5.2 Hardware Requirements

S.No	Hardware Component	Description
1	NVIDIA GPU(A100,RTX)	Performs fast training and references tasks on large medical datasets.
2	TPU	Supports cloud-based execution of Tensor flow based models
3	Jetson Nano / Raspberry Pi	Facilitates local processing of AI models in low-resource or offline healthcare environments
4	Cloud Infrastructure	Hosts APIs, stores reports, and manages compute resources for multiple remote users

both the software and hardware components of the Agentic Fracture Detection system play an indispensable role in ensuring that the solution works smoothly across different environments from high-tech hospitals to small clinics.

CHAPTER 6
SOFTWARE TESTING

Software testing is a crucial phase in the development of the Agentic Fracture Detection (AFD) system. It ensures that the system functions correctly, meets the specified requirements, and provides accurate results. The testing phase involves various methodologies, including unit testing, integration testing, and system testing, to validate the performance of different AI agents within the AFD framework.Thorough testing is essential to detect potential issues early in the development cycle and to improve the reliability of the system. By employing a structured approach to testing, we can ensure that each component of the AFD system works as expected and integrates seamlessly with other components.

6.1. Testing Methodologies

To ensure the reliability and efficiency of the AFD system, multiple testing methodologies have been employed. These methodologies aim to verify the functionality of individual modules, their interactions, and the system's overall performance under real-world conditions.

6.2. Unit Testing

Unit testing focuses on validating individual components of the system. Each AI agent—Perception Agent, Reasoning Agent, Interpretation Agent, and Decision Agent has been tested independently. The Reasoning Agent was tested to verify that it correctly assesses the severity of fractures by analyzing various input features. The Interpretation Agent was evaluated for its ability to generate meaningful and accurate medical reports based on the detected fractures. The Decision Agent was tested for its ability to suggest appropriate treatment recommendations based on severity levels, ensuring that its suggestions align with established medical guidelines.

6.3. Integration Testing

Integration testing ensures that the different agents within the AFD system work together cohesively. This stage of testing focused on verifying the interaction between various agents and ensuring that data flows correctly between them. The interaction between the Perception Agent and the Reasoning Agent was tested to confirm that fracture detection results were correctly passed for severity assessment. Additionally,

we tested whether the Interpretation Agent could generate accurate reports based on the severity assessment provided by the Reasoning Agent. Lastly, we verified that the Decision Agent correctly interpreted severity scores and provided appropriate treatment recommendations.

6.4. System Testing

System testing evaluates the entire AFD framework to verify its performance under real-world conditions. The system was subjected to multiple tests to assess its overall accuracy, efficiency, and robustness.Accuracy testing involved comparing the model's predictions against expert radiologist diagnoses to measure the precision of fracture detection. Performance testing was conducted to measure response time and efficiency when processing large datasets or high-resolution X-ray images. Robustness testing focused on evaluating the system's ability to handle noisy or incomplete X-ray images, ensuring that minor distortions in the data did not significantly impact performance.To carry out effective testing, various tools and frameworks have been utilized to streamline the process and ensure accurate validation of the AFD system.For unit testing of Python-based AI components, PyTest was used to automate test execution and verify expected outputs

Table 6.4 Test Case

Test Case ID	**Description**	**Expected Outcome**
TC-01	Verify Reasoning Agent's fracture severity assessment	Severity score should match expected values
TC-02	Check Interpretation Agent's report generation	Generated reports should align with clinical standards
TC-03	Validate Decision Agent's treatment suggestions	Suggested treatments should be appropriate for severity levels
TC-04	Test system performance on high-resolution X-ray images	System should process images efficiently without lag

The testing phase revealed that the AFD system performs with high accuracy and robustness. The Reasoning Agent demonstrated an accuracy of over 90% in assessing severity levels, ensuring reliable evaluations of fracture seriousness. The Interpretation Agent successfully generated human-readable reports that closely matched those written by radiologists, making it a valuable tool for assisting medical professionals.Furthermore, the Decision Agent's treatment suggestions were reviewed and validated by medical experts, confirming that its recommendations align with standard treatment protocols. The system also performed well in robustness tests, handling variations in image quality without significant loss of accuracy.The software testing process played a crucial role in ensuring that the AFD system not only meets its functional requirements but also adheres to stringent performance standards. By employing a variety of rigorous testing methodologies, the system underwent thorough validation to confirm its accuracy, reliability, and robustness. These comprehensive testing efforts yielded promising results, indicating that the AFD framework has the potential to significantly assist radiologists in their work. It achieves this by providing automated fracture detection, conducting severity assessments, and offering treatment recommendations that are both timely and relevant.

Looking ahead, future testing initiatives will be directed towards enhancing the system's real-time performance, with a particular emphasis on reducing latency. This is essential for ensuring that radiologists can rely on the system for immediate insights during critical decision-making processes. Additionally, there will be a concerted effort to expand the dataset used for testing, which is vital for improving the system's generalization capabilities. By incorporating a wider range of patient demographics and various X-ray imaging conditions, the AFD system aims to become even more effective and adaptable in diverse clinical environments. This ongoing commitment to refinement and enhancement underscores the importance of continuous improvement in medical

CHAPTER 7

FUTURE ENHANCEMENT AND CONCLUSION

7.1 FUTURE ENHANCEMENT

The development of the Agentic Fracture Detection (AFD) system marks a substantial leap forward in applying artificial intelligence to medical diagnostics, particularly in the realm of fracture detection and analysis. Throughout this research, we have explored the critical challenges in traditional fracture identification methods and introduced a novel AI-driven approach that integrates Faster R-CNN with a multi-agent architecture. By leveraging advanced deep learning models and vision-language frameworks, the system enhances diagnostic precision, interpretability, and efficiency, offering essential support to radiologists and healthcare professionals. The findings of this research not only validate the effectiveness of AI-driven diagnostics but also highlight the transformative potential of intelligent systems in modern medical imaging. The advancements introduced in this work lay the foundation for a new era of AI-assisted diagnostic tools, ensuring more precise and reliable fracture detection while reducing the burden on medical professionals.

The AFD system introduces an innovative structured framework, incorporating multiple AI agents, each designated with a specific role in the diagnostic workflow. The Perception Agent ensures accurate fracture detection using state-of-the-art object detection techniques, while the Reasoning Agent evaluates the severity of identified fractures. The Interpretation Agent facilitates human-readable diagnostic reporting, ensuring that AI-generated insights are both comprehensible and actionable for medical professionals. Additionally, the Decision Agent assists in recommending appropriate treatment steps, thereby offering an AI-powered approach to clinical decision-making. This multi-agent system automates the fracture detection process and enhances its reliability, making it a promising addition to modern healthcare environments. The structured approach ensures that radiologists can trust AI-driven insights while maintaining full control over the diagnostic process, ultimately streamlining medical imaging workflows and reducing potential errors in diagnosis.

The experimental results obtained during the implementation and evaluation of the AFD system demonstrate significant improvements in fracture detection accuracy

compared to conventional manual assessments and previous AI-based methodologies. The integration of AI-assisted decision support provides radiologists with highly accurate, consistent, and detailed insights, reducing the risk of misdiagnosis and alleviating the workload in medical imaging departments. By streamlining the workflow in radiology units and emergency care centers, the AFD system has the potential to significantly enhance patient care, accelerate the diagnostic process, and improve clinical efficiency. The system's capacity to operate in diverse clinical settings underscores its robustness and adaptability, making it a valuable asset for healthcare institutions worldwide. Moreover, its ability to integrate seamlessly with existing hospital infrastructure ensures that the adoption of AI-driven diagnostics is both feasible and beneficial in real-world clinical scenarios.

One of the fundamental strengths of this research is its emphasis on interpretability and usability. Traditional AI models often function as black-box systems, creating skepticism among medical professionals regarding their reliability. By incorporating vision-language models and natural language processing techniques, the AFD system bridges the gap between AI-driven diagnostics and human expertise, ensuring that medical professionals can confidently interpret and act upon AI-generated recommendations. This approach fosters greater acceptance of AI in healthcare and sets a strong foundation for future advancements in medical imaging technology. The ability to provide clear, transparent, and context-aware AI explanations enhances trust in AI-powered diagnostics and promotes collaboration between medical experts and intelligent systems. This collaborative approach ensures that the integration of AI does not replace human expertise but rather augments it, leading to more informed and precise medical decision-making.

Despite the promising outcomes achieved, there remain several avenues for future improvements and expansions. Enhancing model generalizability through larger, more diverse, and clinically representative datasets can further refine the system’s performance, ensuring robust detection capabilities across different patient demographics. Additionally, integrating multi-modal imaging techniques such as CT scans and MRIs can provide a more comprehensive diagnostic assessment, improving the detection of complex fractures and associated musculoskeletal conditions. The adoption of federated learning frameworks can also contribute to enhanced data

privacy and security, enabling collaborative AI model training across multiple medical institutions without exposing sensitive patient data. Ensuring that AI models continue to evolve through real-world clinical feedback mechanisms will be essential in achieving sustained accuracy and reliability. By continuously fine-tuning the model using real-world cases and clinical input, the AI system can improve its diagnostic accuracy and maintain its relevance in a rapidly evolving healthcare landscape.

Furthermore, the real-time deployment of the AFD system in clinical settings represents a crucial step forward. Optimizing computational efficiency through techniques such as model compression, quantization, and hardware acceleration can enable seamless integration into hospitals, emergency rooms, and telemedicine platforms. Edge computing and cloud-based implementations can further facilitate remote access, making the system accessible to healthcare professionals in diverse environments, including resource-limited settings. The ability to perform AI-powered diagnostics in real-time will drastically improve response times in critical care scenarios, ensuring that patients receive timely and accurate treatment recommendations. Additionally, integrating AI-powered solutions into mobile health applications and wearable technologies can further expand their accessibility, allowing for early detection and continuous monitoring of musculoskeletal conditions.

Ensuring regulatory compliance and adherence to ethical AI practices will also be vital for large-scale adoption in the healthcare industry. Addressing potential biases in AI predictions, implementing transparent validation processes, and aligning with established medical regulatory standards will foster trust in AI-driven diagnostic solutions. Ethical considerations such as explainability, fairness, and accountability must remain at the forefront of AI development to ensure safe and responsible implementation in real-world medical practice. Future research should continue to explore mechanisms for improving AI transparency and ensuring equitable access to AI-assisted healthcare solutions. It is also crucial to establish clear guidelines for AI .

7.2 Conclusion

In conclusion, this research underscores the profound and transformative impact that artificial intelligence (AI) is having on the field of medical imaging. It is evident that intelligent diagnostic systems, exemplified by the Advanced Fracture Detection (AFD) system, are not merely enhancing the accuracy of fracture detection; they are

revolutionizing the entire diagnostic process. These sophisticated systems are designed to streamline radiological workflows, thereby increasing efficiency and reducing the time required for diagnosis. More importantly, they contribute significantly to improved overall patient outcomes, which is a critical aspect of modern healthcare that cannot be overlooked.

As AI technology continues to evolve at an unprecedented pace, its integration into various facets of healthcare is becoming increasingly essential. This integration will undoubtedly play a pivotal role in shaping the future landscape of diagnostics and treatment planning. The AFD system, in particular, serves as a foundational step toward the development of more advanced, interpretable, and efficient AI-assisted medical tools. This evolution is not just a technological advancement; it is essential for creating a healthcare ecosystem that is not only more effective but also more accessible to a broader range of patients, including those in underserved communities. By continuously improving and expanding its capabilities, the AFD system possesses the potential to revolutionize the process of fracture diagnosis. It sets new benchmarks for AI-driven medical solutions that healthcare professionals can rely upon with confidence. The intersection of AI and healthcare holds immense promise, paving the way for smarter, faster, and more accurate diagnostic methodologies. These advancements will ultimately benefit both medical professionals and patients worldwide, leading to a more informed and responsive healthcare system.

As artificial intelligence (AI) continues to mature and its applications in the field of medicine expand, we can vividly envision a future where technology and human expertise collaborate in a seamless and harmonious manner. This collaboration is poised to not only enhance the quality of patient care and improve diagnostic accuracy but also to lead to significantly better health outcomes and a more efficient healthcare system overall. The ongoing advancements in AI technology are set to refine existing diagnostic processes, making them more precise and reliable, while simultaneously opening new avenues for innovative treatments that were previously unimaginable.

This evolution ensures that the healthcare industry remains at the forefront of scientific progress, continually adapting to meet the ever-evolving needs of patients and healthcare providers alike. In this dynamic landscape, the potential for AI to transform healthcare is not just promising; it is boundless. The integration of AI into healthcare

practices can facilitate personalized medicine, where treatments are tailored to the individual characteristics of each patient, thereby maximizing effectiveness and minimizing adverse effects.

Moreover, AI can assist in predictive analytics, enabling healthcare professionals to anticipate health issues before they arise, thus allowing for proactive interventions that can save lives and reduce costs. It is imperative that we embrace these changes and advancements in technology, as they hold the key to fostering a healthier future for all. By leveraging the capabilities of AI, we can create a healthcare system that is not only more responsive but also more efficient and effective in addressing the diverse needs of patients. The integration of AI into healthcare practices can lead to improved diagnostic accuracy, personalized treatment plans, and enhanced patient engagement. Furthermore, as we continue to harness the power of data and machine learning, we can uncover insights that were previously unattainable, paving the way for innovative solutions to complex health challenges. This transformative approach not only enhances the quality of care but also empowers healthcare providers to make informed decisions that ultimately benefit the entire community.

Appendices

Source code:

YOLO_agent.ipynb

```
import os
import numpy as np
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
import cv2
import random
import matplotlib.image as mpimg
import warnings
warnings.filterwarnings('ignore')
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import *
from tensorflow.keras.callbacks import EarlyStopping , ReduceLROnPlateau
from tensorflow.keras.preprocessing.image import ImageDataGenerator , load_img
from sklearn.metrics import classification_report, confusion_matrix, f1_score
from sklearn.utils import class_weight
from tensorflow.keras.applications import DenseNet121
from keras import backend as keras
from ultralytics import YOLO
from tensorflow.keras.preprocessing.image import load_img, img_to_array
train_images = '/kaggle/input/bone-fracture-dataset-good-data/train/images'
train_labels = '/kaggle/input/bone-fracture-dataset-good-data/train/labels'
test_images = '/kaggle/input/bone-fracture-dataset-good-data/test/images'
test_labels = '/kaggle/input/bone-fracture-dataset-good-data/test/images'
val_images = '/kaggle/input/bone-fracture-dataset-good-data/valid/images'
val_labels = '/kaggle/input/bone-fracture-dataset-good-data/valid/labels'

print('Number of train frames: ' + str(len(os.listdir(train_images))))
```

```
print('Number of train labels: ' + str(len(os.listdir(train_labels))))
print('Number of val frames: ' + str(len(os.listdir(val_images))))
print('Number of val labels: ' + str(len(os.listdir(val_labels))))
print('Number of test frames: ' + str(len(os.listdir(test_images))))
print('Number of test labels: ' + str(len(os.listdir(test_labels))))
print('Total frames: ' + str(len(os.listdir(train_images)) + len(os.listdir(test_images)) +
len(os.listdir(val_images))))
# Get a list of all the image files in the training images directory
image_files = os.listdir(train_images)
# Choose 16 random image files from the list
random_images = random.sample(image_files, 16)
# Set up the plot
fig, axs = plt.subplots(4, 4, figsize=(16, 16))
# Loop over the random images and plot the object detections
for i, image_file in enumerate(random_images):
    row = i // 4
    col = i % 4
    # Load the image
    image_path = os.path.join(train_images, image_file)
    image = cv2.imread(image_path)
    # Load the labels for this image
    label_file = os.path.splitext(image_file)[0] + ".txt"
    label_path = os.path.join(train_labels, label_file)
    with open(label_path, "r") as f:
        labels = f.read().strip().split("\n")
    # Loop over the labels and plot the object detections
    # Loop over the labels and plot the object detections
    for label in labels:
        if len(label.split()) != 5:
            continue
        class_id, x_center, y_center, width, height = map(float, label.split())
        x_min = int((x_center - width/2) * image.shape[1])
```

```
        y_min = int((y_center - height/2) * image.shape[0])
        x_max = int((x_center + width/2) * image.shape[1])
        y_max = int((y_center + height/2) * image.shape[0])
        cv2.rectangle(image, (x_min, y_min), (x_max, y_max), (0, 255, 0), 3)
    # Show the image with the object detections
    axs[row, col].imshow(cv2.cvtColor(image, cv2.COLOR_BGR2RGB))
    axs[row, col].axis('off')
plt.show()
image = cv2.imread('/kaggle/input/bone-fracture-dataset-good-
data/train/images/distal-humeral-fractures-2-
_JPEG_jpg.rf.8a3f89cf8eb406014facd16bfe774d23.jpg')
# Get the size of the image
height, width, channels = image.shape
print(f"The image has dimensions {width}x{height} and {channels} channels.")
# Loading a pretrained model
model = YOLO('yolov8s.pt')
model.train(data = '/kaggle/input/bone-fracture-dataset-good-data/data.yaml',
            epochs = 35,
            imgsz = height,
            seed = 42
)
# read in the results.csv file as a pandas dataframe
df = pd.read_csv('runs/detect/train/results.csv')
df.columns = df.columns.str.strip()
# create subplots using seaborn
fig, axs = plt.subplots(nrows=5, ncols=2, figsize=(15, 15))
# plot the columns using seaborn
sns.lineplot(x='epoch', y='train/box_loss', data=df, ax=axs[0,0])
sns.lineplot(x='epoch', y='train/cls_loss', data=df, ax=axs[0,1])
sns.lineplot(x='epoch', y='train/dfl_loss', data=df, ax=axs[1,0])
sns.lineplot(x='epoch', y='metrics/precision(B)', data=df, ax=axs[1,1])
sns.lineplot(x='epoch', y='metrics/recall(B)', data=df, ax=axs[2,0])
```

```
sns.lineplot(x='epoch', y='metrics/mAP50(B)', data=df, ax=axs[2,1])
sns.lineplot(x='epoch', y='metrics/mAP50-95(B)', data=df, ax=axs[3,0])
sns.lineplot(x='epoch', y='val/box_loss', data=df, ax=axs[3,1])
sns.lineplot(x='epoch', y='val/cls_loss', data=df, ax=axs[4,0])
sns.lineplot(x='epoch', y='val/dfl_loss', data=df, ax=axs[4,1])
# set titles and axis labels for each subplot
axs[0,0].set(title='Train Box Loss')
axs[0,1].set(title='Train Class Loss')
axs[1,0].set(title='Train DFL Loss')
axs[1,1].set(title='Metrics Precision (B)')
axs[2,0].set(title='Metrics Recall (B)')
axs[2,1].set(title='Metrics mAP50 (B)')
axs[3,0].set(title='Metrics mAP50-95 (B)')
axs[3,1].set(title='Validation Box Loss')
axs[4,0].set(title='Validation Class Loss')
axs[4,1].set(title='Validation DFL Loss')
# add suptitle and subheader
plt.suptitle('Training Metrics and Loss', fontsize=24)
# adjust top margin to make space for suptitle
plt.subplots_adjust(top=0.8)
# adjust spacing between subplots
plt.tight_layout()
plt.show()
# Loading the best performing model
model = YOLO('/kaggle/working/runs/detect/train/weights/best.pt')
# Evaluating the model on the test dataset
metrics = model.val(conf = 0.25, split = 'test')
%matplotlib inline
# Create the barplot
ax = sns.barplot(x=['mAP50-95', 'mAP50', 'mAP75'], y=[metrics.box.map,
metrics.box.map50, metrics.box.map75])
```

```
# Set the title and axis labels
ax.set_title('YOLO Evaluation Metrics')
ax.set_xlabel('Metric')
ax.set_ylabel('Value')
# Set the figure size
fig = plt.gcf()
fig.set_size_inches(8, 6)
# Add the values on top of the bars
for p in ax.patches:
        ax.annotate('{:.3f}'.format(p.get_height()),  (p.get_x()  +  p.get_width()  /  2,
p.get_height()), ha='center', va='bottom')
# Show the plot
plt.show()
%matplotlib inline
# Reading the confusion matrix image file
img = mpimg.imread('/kaggle/working/runs/detect/train/confusion_matrix.png')
# Plotting the confusion matrix image
fig, ax = plt.subplots(figsize = (15, 15))
ax.imshow(img)
ax.axis('off');
#Function to perform ship detections
def ship_detect(img_path)
    # Read the image
    img = cv2.imread(img_path)
    # Pass the image through the detection model and get the result
    detect_result = model(img)
    # Plot the detections
    detect_img = detect_result[0].plot()
    # Convert the image to RGB format
    detect_img = cv2.cvtColor(detect_img, cv2.COLOR_BGR2RGB)
    return detect_img
```

```
import random
# Define the directory where the custom images are stored
custom_image_dir = test_images
# Get the list of image files in the directory
image_files = os.listdir(custom_image_dir)
# Select 16 random images from the list
selected_images = random.sample(image_files, 16)
# Create a figure with subplots for each image
fig, axes = plt.subplots(nrows=4, ncols=4, figsize=(15, 15))
# Iterate over the selected images and plot each one
for i, img_file in enumerate(selected_images):
    row_idx = i // 4
    col_idx = i % 4
    # Load the current image and run object detection
    img_path = os.path.join(custom_image_dir, img_file)
    detect_img = ship_detect(img_path)
    # Plot the current image on the appropriate subplot
    axes[row_idx, col_idx].imshow(detect_img)
    axes[row_idx, col_idx].axis('off')
```

Faster_ R_CNN.ipynb

```
import numpy as np # linear algebra
import pandas as pd # data processing, CSV file I/O (e.g. pd.read_csv)
import os
import torch
from torch import nn
import albumentations as A
# import torchsummary
import cv2
import matplotlib.pyplot as plt
import seaborn as sns
from tqdm import tqdm
import random
```

```
torch.cuda.empty_cache()
BS=16
LR=0.00005
epochs=20
IS=256
D='cuda' if torch.cuda.is_available() else 'cpu'
num_classes = 7
classes=['elbow positive', 'fingers positive', 'forearm fracture', 'humerus fracture', 'humerus', 'shoulder fracture', 'wrist positive']
c2l={k:v for k,v in list(zip(classes,list(range(num_classes))))}
l2c={v:k for k,v in c2l.items()}
dir_path='/kaggle/input/bone-fracture-detection-computer-vision-project/bone fracture detection.v4-v4.yolov8'
train_dir_path='/kaggle/input/bone-fracture-detection-computer-vision-project/bone fracture detection.v4-v4.yolov8/train'
train_img_paths = sorted(os.listdir('/kaggle/input/bone-fracture-detection-computer-vision-project/bone fracture detection.v4-v4.yolov8/train/images'))
train_target_paths = sorted(os.listdir('/kaggle/input/bone-fracture-detection-computer-vision-project/bone fracture detection.v4-v4.yolov8/train/labels'))
val_dir_path='/kaggle/input/bone-fracture-detection-computer-vision-project/bone fracture detection.v4-v4.yolov8/valid'
val_img_paths = sorted(os.listdir('/kaggle/input/bone-fracture-detection-computer-vision-project/bone fracture detection.v4-v4.yolov8/valid/images'))
val_target_paths = sorted(os.listdir('/kaggle/input/bone-fracture-detection-computer-vision-project/bone fracture detection.v4-v4.yolov8/valid/labels'))
def unconvert(width, height, x, y, w, h):
    xmax = int((x*width) + (w * width)/2.0)
    xmin = int((x*width) - (w * width)/2.0)
    ymax = int((y*height) + (h * height)/2.0)
    ymin = int((y*height) - (h * height)/2.0)
    return xmin, ymin, xmax, ymax
idx=random.randint(0,3000)
```

```
# idx=0
ip=os.path.join(train_dir_path,'images',train_img_paths[idx])
tp=os.path.join(train_dir_path,'labels',train_target_paths[idx])
image=cv2.imread(ip)
image=cv2.cvtColor(image,cv2.COLOR_BGR2RGB)
image=cv2.resize(image,(320,320)
file=open(tp,'r')
target=list(map(float,file.read().split()))[1:]
a=0
while a<len(target):
    bbox=target[a:a+4]
    if len(bbox)==4:
        x,y,w,h=bbox[0], bbox[1], bbox[2], bbox[3]
        xmin,ymin,xmax,ymax=unconvert(320,320,x,y,w,h)
        sp,ep=(xmin,ymin),(xmax,ymax)
        image=cv2.rectangle(image,sp,ep,(255,0,0),2)
    a+=4
plt.imshow(image)
plt.show()
augs=A.Compose([
    A.Resize(IS,IS),
],bbox_params=A.BboxParams(format='pascal_voc',      label_fields=['class_labels']),
is_check_shapes=True)
class FractureData(torch.utils.data.Dataset):
    def __init__(self, dir_path, img_paths, target_paths, augs=None):
        self.dir_path=dir_path
        self.img_paths=img_paths
        self.target_paths=target_paths
        self.augs=augs
    def __len__(self):
        return len(self.img_paths)
    def __getitem__(self,idx):
```

```
    ip=os.path.join(self.dir_path,'images',self.img_paths[idx])
    tp=os.path.join(self.dir_path,'labels',self.target_paths[idx])
    image=cv2.imread(ip)
    image=cv2.cvtColor(image,cv2.COLOR_BGR2RGB)
    H,W,_=image.shap
    file=open(tp,'r')
    target=list(map(float,file.read().split())
    try:
        label=[target.pop(0)]
        bbox=[]
        i=0
        while i<len(target):
            x,y,w,h=target[i:i+4]
            bbox.append([*unconvert(W,H,x,y,w,h)])
            i+=4
        label=label*len(bbox)
        if self.augs!=None:
          data=self.augs(image=image,bboxes=bbox,class_labels=['None']*len(label))
            image=data['image']
            bbox=data['bboxes']
    except:
        if idx+1<len(self.img_paths):
            return self.__getitem__(idx+1)
        else:
            return self.__getitem__(0)
    image=torch.Tensor(np.transpose(image,(2,0,1)))/255.0
    bbox=torch.Tensor(bbox).long()
    label=torch.Tensor(label).long()
    annot={'boxes':bbox,'labels':label}
    return image, annot
def collate_fn(self,batch):
    return tuple(zip(*batch))
```

```
trainset=FractureData(train_dir_path, train_img_paths, train_target_paths, augs)
valset=FractureData(val_dir_path, val_img_paths, val_target_paths, augs)
trainloader=torch.utils.data.DataLoader(trainset,batch_size=BS,collate_fn=trainset.c
ollate_fn)
valloader=torch.utils.data.DataLoader(valset,batch_size=BS,collate_fn=valset.collate
_fn)
print(f'Training Data:- {len(trainset)} images divided into {len(trainloader)} batches')
for image,target in trainloader:
    break
idx=random.randint(0,len(image))
img, tar = image[idx].permute(1,2,0).numpy(), target[idx]
for bbox in tar['boxes']:
        xmin,ymin,xmax,ymax=bbox[0].item(),  bbox[1].item(),  bbox[2].item(),
bbox[3].item()
    sp,ep=(xmin,ymin),(xmax,ymax)
    cv2.rectangle(img,sp,ep,(0,255,0),1)
plt.imshow(img)
plt.title(l2c[tar['labels'][0].item()])
plt.show()
import torchvision
from torchvision.models.detection.faster_rcnn import FastRCNNPredictor
model=torchvision.models.detection.fasterrcnn_resnet50_fpn(preTrained=True)
in_features = model.roi_heads.box_predictor.cls_score.in_features
model.roi_heads.box_predictor = FastRCNNPredictor(in_features, num_classes)
model.to(D);
def trainarc(model, dataloader, opt):
    model.train()
    train_loss=0.0
    for images, targets in tqdm(dataloader):
        image=[i.to(D) for i in images]
        target=[{k:v.to(D) for k,v in ele.items()} for ele in targets]
        opt.zero_grad()
```

```
        losses=model(image,target)
        loss=sum(loss for loss in losses.values())
        loss.backward()
        opt.step()
        train_loss+=loss
    return train_loss/len(dataloader)
def evalarc(model, dataloader):
    model.train()
    val_loss=0.0
    with torch.no_grad():
        for images, targets in tqdm(dataloader):
            image=[i.to(D) for i in images]
            target=[{k:v.to(D) for k,v in ele.items()} for ele in targets]
            losses=model(image,target)
            loss=sum( loss for loss in losses.values() )
            val_loss+=loss
    return val_loss/len(dataloader)
best_val_loss=np.Inf
opt=torch.optim.Adam(model.parameters(),lr=LR)
for i in range(epochs):
    train_loss=trainarc(model,trainloader,opt)
    val_loss=evalarc(model,valloader)
    print(f"Epochs: {i+1}/{epochs}:- Trainloss: {train_loss}, Valloss: {val_loss}")
    if val_loss<best_val_loss:
        torch.save(model.state_dict(),'/kaggle/working/model.pt')
        print("Model Updated")
        best_val_loss=val_loss

torch.save(model.state_dict(),'/kaggle/working/FullyTrainedModel.pt')
print("Fully Trained Model Saved")
print(f"Done. Best Val Loss: {best_val_loss}")
```

```
test_dir_path='/kaggle/input/bone-fracture-detection-computer-vision-project/bone fracture detection.v4-v4.yolov8/test'
test_img_paths = sorted(os.listdir('/kaggle/input/bone-fracture-detection-computer-vision-project/bone fracture detection.v4-v4.yolov8/test/images'))
test_target_paths = sorted(os.listdir('/kaggle/input/bone-fracture-detection-computer-vision-project/bone fracture detection.v4-v4.yolov8/test/labels'))
testset=FractureData(test_dir_path, test_img_paths, test_target_paths, augs)
idx=random.randint(0,len(testset)-1)
print(idx)
test_img,test_tar=testset[idx]
model.eval()
pred=model(test_img.unsqueeze(0).to(D))
# pred=torchvision.ops.nms(pred[0]['boxes'].detach(),pred[0]['scores'].detach(),0.02)
xmin,ymin,xmax,ymax=pred[0]['boxes'][0].detach().cpu().long().tolist()
label=pred[0]['labels'][0].item()
Txmin,Tymin,Txmax,Tymax=test_tar['boxes'][0].tolist()
image=cv2.rectangle(test_img.permute(1,2,0).numpy(),(xmin,ymin),(xmax,ymax),(255,0,0),2)
image=cv2.rectangle(image,(Txmin,Tymin),(Txmax,Tymax),(0,255,0),2)
plt.imshow(image)
plt.title(l2c[label])
plt.show()
```

inference.py

```
import torch
import torchvision.transforms as T
from PIL import Image, ImageDraw
import torchvision
from agent1 import VLMagent
class FractureSegmentation:
    def __init__(self):
        self.model = self.load_model("model.pt")
```

```
    def load_model(self,model_path, num_classes=7):  # Set the correct number of
classes
        # Load a base Faster R-CNN model (without pre-trained COCO head)
        model = torchvision.models.detection.fasterrcnn_resnet50_fpn(pretrained=Fals
        # Modify the classification head to match the number of classes
        in_features = model.roi_heads.box_predictor.cls_score.in_features
                                                    model.roi_heads.box_predictor =
torchvision.models.detection.faster_rcnn.FastRCNNPredictor(in_features,
num_classes)
        # Load the trained weights
        state_dict = torch.load(model_path, map_location=torch.device('cpu'))
        model.load_state_dict(state_dict)
       model.eval()
        return model
        def preprocess_image(self,image):
            transform = T.Compose([
           T.ToTensor(),
        ])
        return transform(image).unsqueeze(0), image
    def draw_bounding_boxes(self,image, boxes, labels, scores, threshold=0.5):
        draw = ImageDraw.Draw(image)
        for box, label, score in zip(boxes, labels, scores):
            if score >= threshold:
                draw.rectangle(box.tolist(), outline="red", width=3)
                draw.text((box[0], box[1]), f"{label}: {score:.2f}", fill="red")
        return image
    def Analyse(self,image_path = "Sample_data/image.jpg", threshold=0.5):
        image_tensor, image = self.preprocess_image(image_path)
        with torch.no_grad():
            predictions = self.model(image_tensor)
        boxes = predictions[0]['boxes']
        labels = predictions[0]['labels']
```

```
        scores = predictions[0]['scores']
            output_image = self.draw_bounding_boxes(image, boxes, labels, scores, threshold)
        agent = VLMagent()
        response = agent.vlm(output_image)
        return output_image,response
```

agents.py

```
import torch
from transformers import AutoModelForSeq2SeqLM, AutoTokenizer

prompt = """
    **Role:** You are an advanced AI system designed for fracture assessment in medical imaging. Your goal is to analyze fracture severity, interpret findings into human-readable reports, and provide decision support for diagnosis and treatment.

    **Task:** Given an X-ray or CT scan image, you will function as a multi-agent system with the following responsibilities:
    1. I already detected fracture with other model and make an boundary box. if boundary box is not present in image consider it is a fracture free and fill output formate all null
    1. **Reasoning Agent** – Assess the severity of the detected fracture based on medical imaging features.
    2. **Interpretation Agent** – Generate a detailed, structured report using a vision-language model, ensuring clarity for radiologists and healthcare professionals.
    3. **Decision Agent** – Suggest appropriate next steps for diagnosis, additional tests, and potential treatment recommendations.

    **Instructions:**
    - **Fracture Detection:** Identify and classify fractures based on severity (e.g., no fracture , mild, moderate, severe).
```

- **Report Generation:** Provide a structured radiology report, including location, type of fracture, and any relevant observations. if no fracture say healthy.

- **Decision Support:** Offer evidence-based recommendations for follow-up imaging, specialist consultations, or immediate interventions.

- **Explainability:** Ensure outputs are transparent, citing key imaging features that influenced your assessment.

Output Format:

- **Fracture Severity:** [Classification]

- **Findings:** [Concise description of fracture and affected area]

- **Suggested Next Steps:** [Diagnosis & treatment recommendations]

Your responses should be accurate, clear, and aligned with medical best practices.

you responses should be in Output format . no other word or sentences .

"""

```
# 1. Reasoning Agent - Evaluates the severity of fractures
class ReasoningAgent:
    def __init__(self, model_path='fracture_severity_model.pth'):
        self.model = self.load_model(model_path)
    def load_model(self, model_path):
        model = torch.nn.Linear(10, 1)  # Placeholder model
        model.load_state_dict(torch.load(model_path))
        model.eval()
        return model
    def assess_severity(self, features):
        """Predicts fracture severity on a scale of 0 to 1"""
        features = torch.tensor(features).float()
        severity_score = self.model(features).item()
        return severity_score
# 2. Interpretation Agent - Generates a readable medical report
class InterpretationAgent:
    def __init__(self, model_name='facebook/bart-large-cnn'):
        self.model = AutoModelForSeq2SeqLM.from_pretrained(model_name)
```

```
        self.tokenizer = AutoTokenizer.from_pretrained(model_name)
    def generate_report(self, severity, fracture_type):
        prompt = f"The patient has a {fracture_type} fracture with a severity score of {severity}. Generate a medical summary."
        inputs = self.tokenizer(prompt, return_tensors='pt', max_length=512, truncation=True)
        summary_ids = self.model.generate(**inputs, max_length=150)
        return self.tokenizer.decode(summary_ids[0], skip_special_tokens=True)
# 3. Decision Agent - Suggests next steps for treatment
class DecisionAgent:
    def suggest_treatment(self, severity, fracture_type):
        if severity < 0.3:
            return f"The {fracture_type} fracture is mild. Suggested treatment: Pain management and immobilization."
        elif 0.3 <= severity < 0.7:
            return f"The {fracture_type} fracture is moderate. Suggcested treatment: X-ray follow-up and possible casting."
        else:
            return f"The {fracture_type} fracture is severe. Suggested treatment: Immediate surgical intervention."
# Example Execution
reasoning_agent = ReasoningAgent()
interpretation_agent = InterpretationAgent()
decision_agent = DecisionAgent()
# Sample features (placeholder values)
features = [0.5] * 10  # Example input for the Reasoning Agent
severity = reasoning_agent.assess_severity(features)
fracture_type = "transverse"
report = interpretation_agent.generate_report(severity, fracture_type)
decision = decision_agent.suggest_treatment(severity, fracture_type)
print("Generated Report:\n", report)
print("Decision Support:\n", decision)
```

app.py

```
import streamlit as st
from inferance import FractureSegmentation
from PIL import Image
st.title("Fracture Segmentation and Analysis")
uploaded_file = st.file_uploader("Upload an X-ray image", type=["png", "jpg", "jpeg"])
if uploaded_file is not None:
    image = Image.open(uploaded_file)
    st.image(image, caption="Uploaded X-ray Image", use_container_width=True)
    # Process the image
    fs = FractureSegmentation()
    boxed_image, diagnosis = fs.Analyse(image)
    st.subheader("Fracture Segmentation")
    # Display the processed image
            st.image(boxed_image, caption="Fracture Detected Image", use_container_width=True)
    # Display the diagnosis text
    st.subheader("Diagnosis")
st.write(diagnosis)
```

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